Week 15

Course Level Objectives:

- Describe and compute confidence intervals. (CLO 6)
- Solve linear regression and correlation problems. (CLO 7)
- Perform hypothesis testing using statistical methods. (CLO 8)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Identify when ANOVA is appropriate. (CLO 8) **2.** State H_0 and H_a for ANOVA in symbols and in words. (CLO 8) **3.** Explain what SSG and SSE measure. (CLO 8) **4.** Explain how the ratio of between-group to within-group variability gives evidence against H_0. (CLO 8) **5.** Complete an ANOVA table (df, MS, F) from given sums of squares. (CLO 8) **6.** Interpret the F statistic and p-value in context. (CLO 8) **7.** Check the conditions for ANOVA. (CLO 8)	e-Text: 8.1 Analysis of Variance Interactive Lecture: 8.1 Analysis of Variance Slides: 8.1 Analysis of Variance StatKey – Theoretical Distributions (F)	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide; Your Turn: Complete the ANOVA Table (fertilizer and seedling growth)	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom
8. State the simple linear model. (CLO 7) **9.** Distinguish the population slope β_1 and correlation ρ from the sample estimates b_1 and r. (CLO 7) **10.** Interpret the slope in context. (CLO 7) **11.** Use regression output to test H_0: $\beta_1 = 0$ with a t statistic on $n - 2$ degrees of freedom. (CLO 7, CLO 8) **12.** Explain why the tests for slope and for correlation give the same p-value. (CLO 7) **13.** Construct and interpret a confidence interval for the slope. (CLO 6, CLO 7) **14.** Interpret R^2 as the proportion of variability in the response explained by the model. (CLO 7) **15.** Check the conditions for regression inference with a residual plot. (CLO 7)	e-Text: 9.1 Inference for Slope and Correlation Interactive Lecture: 9.1 Inference for Slope and Correlation Slides: 9.1 Inference for Slope and Correlation	Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, R^2); Your Turn: Alcohol and Missed Classes (drinks per week vs. classes missed)	Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom